

MTH166

Lecture-5

Equs. of First Order and Higher Degree

Topic:

Equations of First Order and Higher Degree

Learning Outcomes:

1. What are the Equations of First Order and Higher Degree.
2. How to solve these equations first order and higher degree.

Equations of First Order and Higher Degree:

Let $\frac{dy}{dx} = p$ (A standard notation)

$$\Rightarrow \left(\frac{dy}{dx}\right)^2 = p^2 \quad \left(\text{Not } \frac{d^2y}{dx^2}\right)$$

$$\Rightarrow \left(\frac{dy}{dx}\right)^3 = p^3 \quad \left(\text{Not } \frac{d^3y}{dx^3}\right)$$

$$\Rightarrow \left(\frac{dy}{dx}\right)^4 = p^4 \text{ and so on...}$$

These types of equations can also be written as: $f(x, y, p) = 0$

Equations Solvable for p:

Find the general solution of following differential equations:

Problem 1 $y \left(\frac{dy}{dx} \right)^2 + (x - y) \left(\frac{dy}{dx} \right) - x = 0$ (1)

Solution: Let $\frac{dy}{dx} = p$

Equation (1) can be re-written as:

$$yp^2 + (x - y)p - x = 0$$

$$\Rightarrow yp^2 + px - py - x = 0$$

$$\Rightarrow p(yp + x) - 1(yp + x) = 0$$

$$\Rightarrow (yp + x)(p - 1) = 0$$

Here $(yp + x) = 0$

$$\Rightarrow y \frac{dy}{dx} + x = 0$$

$$\Rightarrow ydy + xdx = 0$$

$$\Rightarrow \int ydy + \int xdx = c_1$$

$$\Rightarrow \frac{y^2}{2} + \frac{x^2}{2} = c_1$$

$$\Rightarrow (y^2 + x^2 - 2c_1) = 0$$

Also $(p - 1) = 0$

$$\Rightarrow \frac{dy}{dx} - 1 = 0$$

$$\Rightarrow dy - dx = 0$$

$$\Rightarrow \int dy - \int dx = 0$$

$$\Rightarrow y - x = c_2$$

$$\Rightarrow (y - x - c_2) = 0$$

So, the general solution of equation (1) is given by:

$$(y^2 + x^2 - 2c_1)(y - x - c_2) = 0 \text{ **Answer.**}$$

Problem 2. $xy \left(\frac{dy}{dx}\right)^2 - (x^2 + y^2) \left(\frac{dy}{dx}\right) + xy = 0$ (1)

Solution: Let $\frac{dy}{dx} = p$

Equation (1) can be re-written as:

$$\begin{aligned} xyp^2 - (x^2 + y^2)p + xy &= 0 \\ \Rightarrow xyp^2 - x^2p - y^2p + xy &= 0 \\ \Rightarrow xp yp - x^2 - y^2p + xy &= 0 \\ \Rightarrow (yp - x)(xp - y) &= 0 \end{aligned}$$

Here $(yp - x) = 0$

$$\begin{aligned} \Rightarrow y \frac{dy}{dx} - x &= 0 \\ \Rightarrow y dy - x dx &= 0 \\ \Rightarrow \int y dy - \int x dx &= c_1 \\ \Rightarrow \frac{y^2}{2} - \frac{x^2}{2} &= c_1 \\ \Rightarrow (y^2 - x^2 - 2c_1) &= 0 \end{aligned}$$

$$\text{Here } (xp - y) = 0$$

$$\Rightarrow x \frac{dy}{dx} - y = 0$$

$$\Rightarrow xdy - ydx = 0$$

$$\Rightarrow xdy = ydx$$

$$\Rightarrow \frac{dy}{y} = \frac{dx}{x}$$

$$\Rightarrow \int \frac{dy}{y} = \int \frac{dx}{x}$$

$$\Rightarrow \log y = \log x + \log c_2$$

$$\Rightarrow \log y = \log xc_2$$

$$\Rightarrow y = xc_2 \quad \Rightarrow (y - xc_2) = 0$$

So, the general solution of equation (1) is given by:

$$(y^2 - x^2 - 2c_1)(y - xc_2) = 0 \text{ **Answer.**}$$

Polling Quiz

What is the relationship between a and b , so that the given equation:

$(x^8 + x + ay^2)dx + (y^8 - y + bxy)dy = 0$ is exact:

(A) $b = 2a$ (Answer)

(B) $a = b$

(C) $a \neq b$

(D) $a = 1, b = 3$

Problem 3. $p^3 + 2xp^2 - y^2p^2 - 2xy^2p = 0$ (1)

Solution: $p^3 + 2xp^2 - y^2p^2 - 2xy^2p = 0$ (1)

$$\Rightarrow p[p^2 + 2xp - y^2p - 2xy^2] = 0$$

$$\Rightarrow p[p(p + 2x) - y^2(p + 2x)] = 0$$

$$\Rightarrow p(p + 2x)(p - y^2) = 0$$

$$\text{Here } p = 0 \quad \Rightarrow \frac{dy}{dx} = 0 \quad \Rightarrow y = c_1 \quad \Rightarrow (y - c_1) = 0$$

$$\text{Also } (p + 2x) = 0$$

$$\Rightarrow \frac{dy}{dx} + 2x = 0$$

$$\Rightarrow dy + 2xdx = 0 \quad \Rightarrow \int dy + 2 \int xdx = c_2 \quad \Rightarrow y + 2\left(\frac{x^2}{2}\right) = c_2$$

$$\Rightarrow (y + x^2 - c_2) = 0$$

$$\text{Also } (p - y^2) = 0$$

$$\Rightarrow \frac{dy}{dx} - y^2 = 0$$

$$\Rightarrow \frac{dy}{dx} = y^2$$

$$\Rightarrow \frac{dy}{y^2} = dx$$

$$\Rightarrow \int y^{-2} dy = \int dx$$

$$\Rightarrow \frac{-1}{y} = x + c_3$$

$$\Rightarrow (xy + 1 + yc_3) = 0$$

So, the general solution of equation (1) is given by:

$$(y - c_1)(y + x^2 - c_2)(xy + 1 + yc_3) = 0 \text{ **Answer.**}$$

Problem 4. $p(p + y) = x(x + y)$ (1)

Solution: $p(p + y) = x(x + y)$ (1)

$$\Rightarrow p^2 + yp - x^2 - xy = 0$$

$$\Rightarrow (p^2 - x^2) + y(p - x) = 0$$

$$\Rightarrow (p - x)(p + x) + y(p - x) = 0$$

$$\Rightarrow (p - x)(p + x + y) = 0$$

Here $(p - x) = 0$

$$\Rightarrow \frac{dy}{dx} - x = 0$$

$$\Rightarrow dy - xdx = 0 \quad \Rightarrow \int dy - \int xdx = c_1 \quad \Rightarrow y - \left(\frac{x^2}{2}\right) = c_1$$

$$\Rightarrow (2y - x^2 - 2c_1) = 0$$

$$\text{Also } (p + x + y) = 0$$

$$\Rightarrow \frac{dy}{dx} + x + y = 0$$

$$\Rightarrow (x + y)dx + dy = 0 \quad (2)$$

Comparing it with: $Mdx + Ndy = 0$

$$\text{Here } M = (x + y) \Rightarrow \frac{\partial M}{\partial y} = 1$$

$$\text{and } N = 1 \Rightarrow \frac{\partial N}{\partial x} = 0$$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, So, equation (2) is non-exact.

Here $\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N} = \frac{1-0}{1} = 1 = x^0 = f(x)$ which can be considered as a function of x .

$$\text{So, I.F.} = e^{\int f(x)dx} = e^{\int 1dx} = e^x$$

Multiplying equation (2) by I.F.

$$[(x + y)dx + dy] \times e^x = 0$$

$$\Rightarrow (xe^x + ye^x)dx + e^x dy = 0$$

Which is an exact differential equation.

$$\text{Solution: } \int_{y=\text{con.}} Mdx + \int (\text{Terms of } N \text{ free from } x)dy = c_2$$

$$\Rightarrow \int_{y=\text{con.}} (xe^x + ye^x)dx + 0 = c_2$$

$$\Rightarrow e^x(x - 1) + ye^x = c_2$$

$$\Rightarrow (e^x(x - 1) + ye^x - c_2) = 0$$

So, the general solution of equation (1) is given by:

$$(2y - x^2 - 2c_1)(e^x(x - 1) + ye^x - c_2) = 0 \text{ **Answer.**}$$

Polling Quiz

In the non-exact differential equation: $M(x, y)dx + N(x, y)dy = 0$

if:

$\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N} = f(x)$ then Integrating factor is given by:

(A) $f(x)$

(B) $\log f(x)$

(C) $e^{f(x)}$

(D) $e^{\int f(x)dx}$ (Answer)

Problem 5. $y = x\left[p + \sqrt{1 + p^2}\right]$

Try it yourself.

Problem 6. $\frac{dy}{dx} - \frac{dx}{dy} = \frac{x}{y} - \frac{y}{x}$

Try it yourself.

